

Data for Policy Analysis and Management

CLASS 7: DATA
VISUALIZATION IN R AND
R MARKDOWN

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Summer 2026

Agenda

1. Why Data Visualization Matters
2. Introduction to ggplot2
3. Communication
4. Introduction to Rmarkdown
5. Lab: Data visualization and Rmarkdown

Why Data Visualization Matters

Turning Data into Information

- Visualization transforms raw data into meaningful insights, helping researchers and policymakers understand complex information at a glance.

Key Purposes of Visualization

- Explore patterns and relationships in data.
- Identify trends, outliers, and unusual observations.
- Communicate complex findings clearly.
- Translate statistical results into meaningful information for broader audiences.

CLASS 6



The Role of Visualization in the Research Process

1

Data Exploration:

- Understand distributions and variation.
- Identify missing data and errors.
- Examine variable relationships.

2

Data Analysis:

- Assess model assumptions.
- Compare groups and identify differences.
- Explore explanations for findings.

3

Communication of Results:

- Present results clearly.
- Highlight important findings.
- Make research accessible to diverse audiences.

Conclusion

Visualizations link statistical evidence to substantive meaning



Introduction to ggplot2

A Framework for Creating Visualizations in R

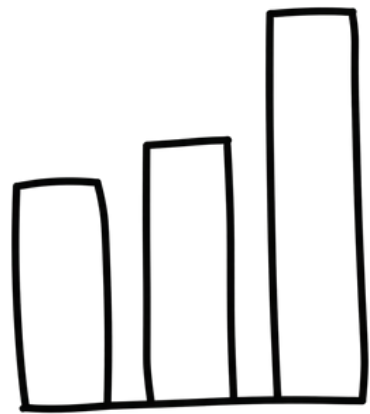
ggplot2 is built on the Grammar of Graphics, a systematic approach that combines data, variables, visual representations, and design choices into a coherent framework for thinking about visualization.

Key Features

- Creates professional-quality graphics suitable for research and publication.
- Consistent approach applicable to a wide range of visualization types.
- Builds both simple and complex figures with the same logic.
- Integrates seamlessly with the tidyverse ecosystem.

<https://ggplot2.tidyverse.org/articles/ggplot2.html>

Choosing the Right Visualization



Comparing Groups

- Wage differences by provider type; demographic outcomes across regions.
- Bar charts, boxplots, grouped comparisons



Examining Relationships

- Experience vs wages; resource-outcome associations across providers.
- Scatterplots, trend lines.
- Understanding Distributions.
- Wage distribution; provider market duration; enrollment spread.
- Histograms, density plots.

<https://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html>

Visualizing Relationships and Patterns

Going Beyond Simple Descriptions

- Visualization reveals what tables and models alone cannot show: uncovering structure, patterns, and meaning hidden within data.

Visualization Helps Examine

- Relationships between variables: how one factor connects to another.
- Differences across groups: comparing outcomes by provider type, region, or policy
- Changes over time: tracking trends before and after interventions.
- Unexpected observations: outliers and anomalies that prompt new questions.
- Hypothesis generation: guiding what to model and how to interpret results.



Designing Effective Data Visualizations



Clarity

- Use clear labels and descriptive titles.
- Highlight the main message so readers immediately understand what the figure communicates.



Accuracy & Simplicity

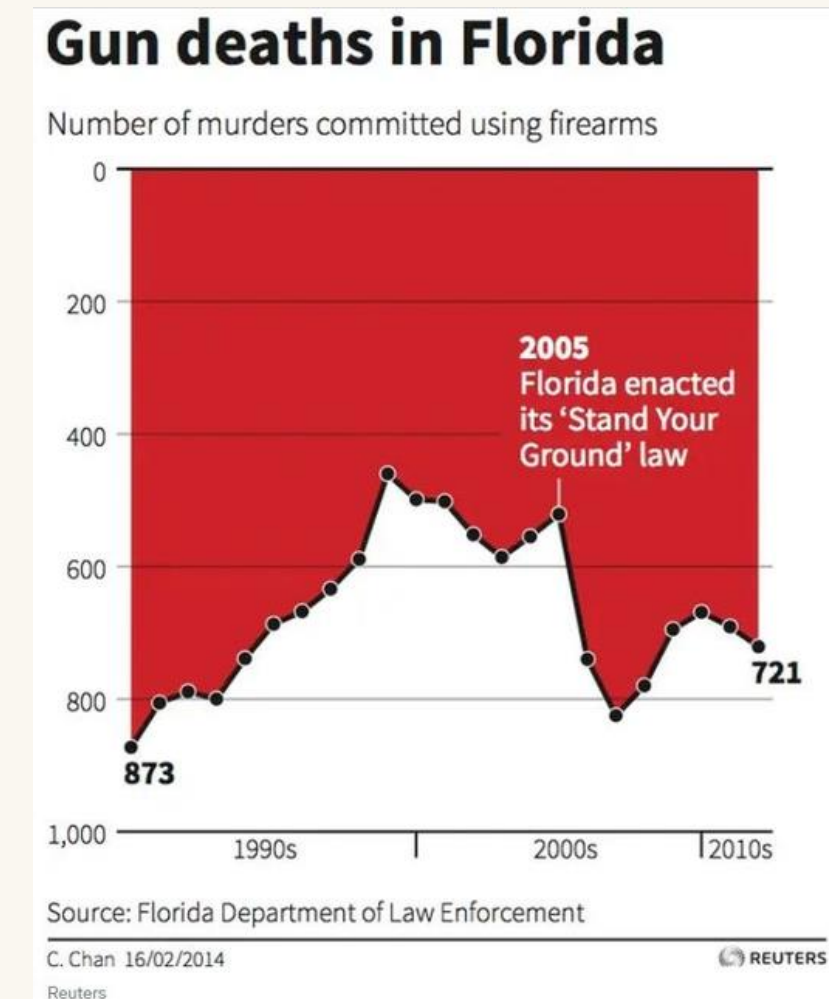
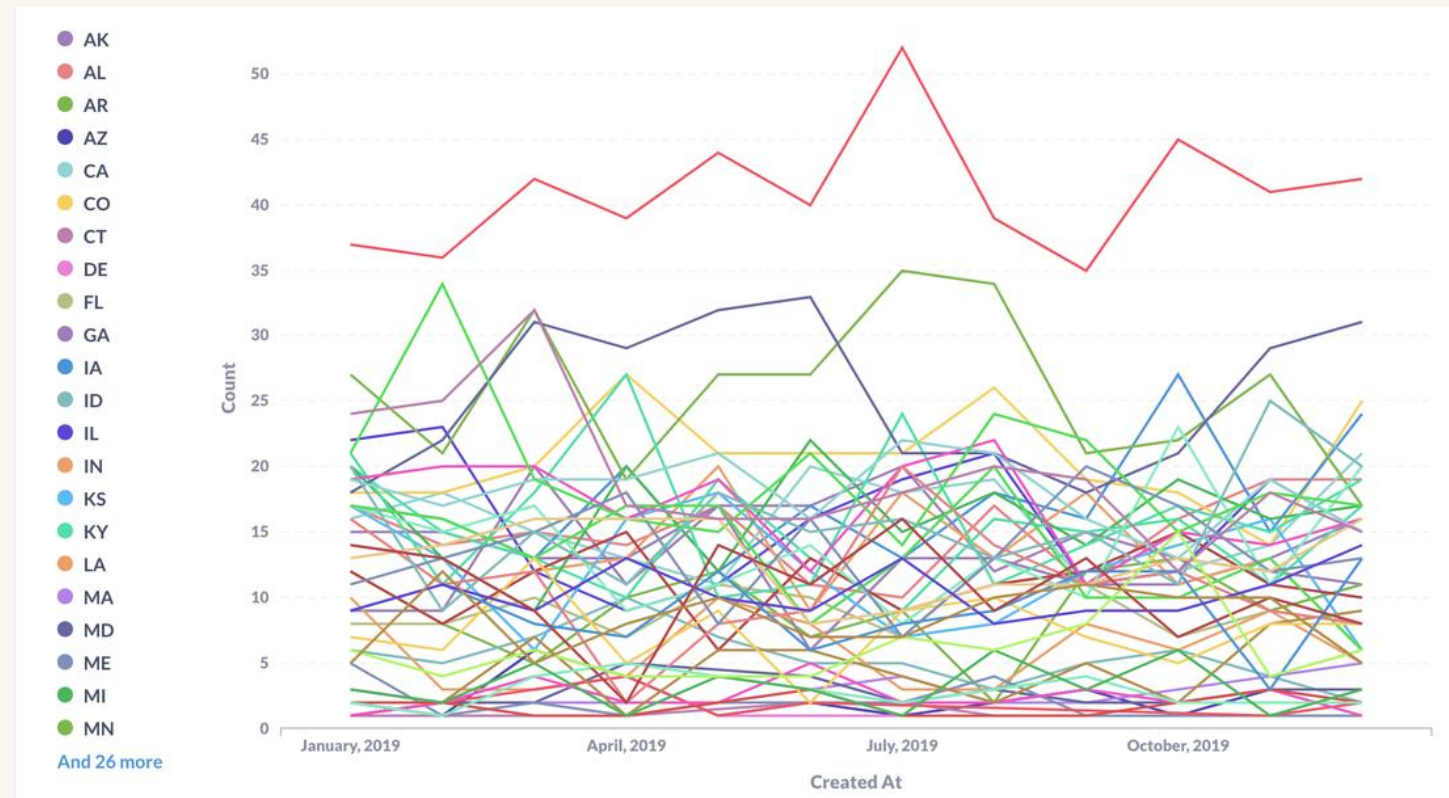
- Avoid misleading scales and represent data honestly.
- Remove clutter and unnecessary details that distract from the core finding.



Consistency

- Apply uniform styles, colors, and formats across all figures to ease comparisons and reduce cognitive load for readers.

Common Visualization Challenges



Too Much Information

- Overloading a figure with too many categories, labels, or data series makes it hard to read.
- Poor visualization choices can hide patterns rather than reveal them.
- Simplicity and focus are essential for effective communication.

Misleading Presentations

- Manipulating scales, truncating axes, or exaggerating differences distorts the evidence. Ignoring context leaves audiences without the information needed to interpret findings.
- The goal is to communicate evidence clearly and honestly, not to impress visually.

Moving From Analysis to Communication

Researchers often need to communicate findings to audiences with different levels of statistical knowledge.

Effective visualizations help:

- Policymakers understand trends and impacts.
- Organizations make decisions based on evidence.
- Communities engage with research findings.
- Researchers communicate results more effectively.

Examples:

- Maps showing geographic patterns.
- Trends showing changes over time.
- Comparisons showing differences across groups.



Introduction to R Markdown

R Markdown combines written explanations, data analysis, statistical results, tables, and figures into a single document, enabling fully reproducible research.

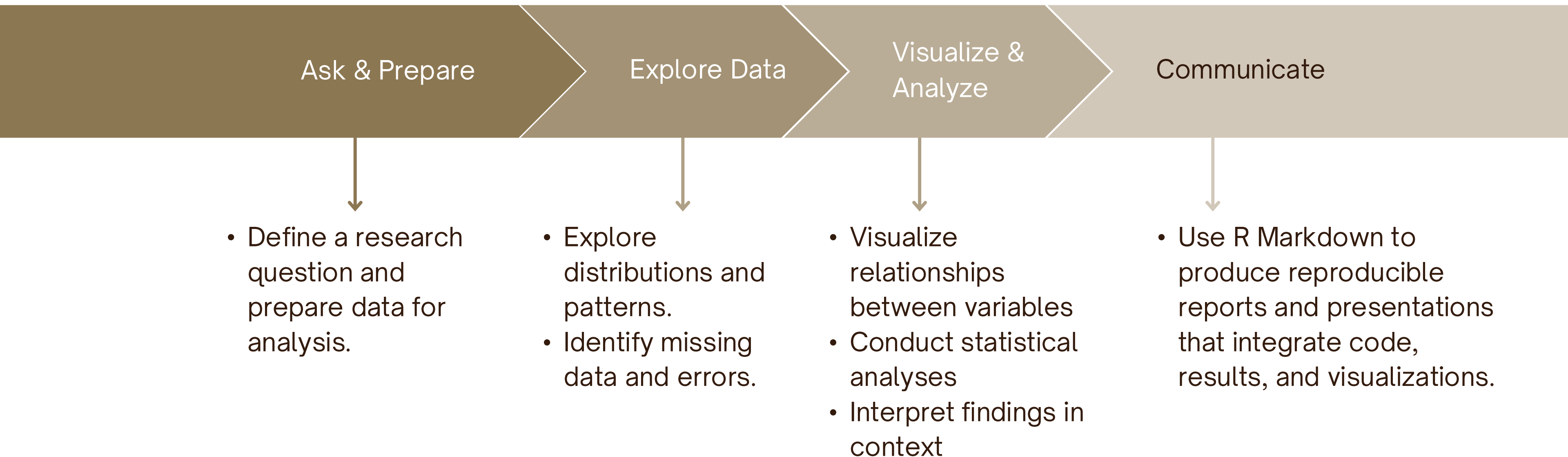
Why Reproducibility Matters:

- Documents the full analytical process.
- Facilitates updates when new data is available.
- Reduces manual errors in reporting.
- Promotes transparent and open research.

R Markdown supports research that is understandable, transparent, and reproducible.



Integrating Visualization and R Markdown



Next Class

- Introduction to Experimental and Quasi-experimental methods
- Research Article Presentations - Grade rubric (10 points total):

Criterion	Points	Explanation
Study Goals & Research Questions	2	Clearly explains the purpose of the study and accurately identifies the research question(s) or hypotheses.
Explanation of Quantitative Methodology	3	Provides a clear, accurate, and accessible explanation of the quantitative methods, including what the authors did and why they chose those methods.
Critical Evaluation of Methodology	2	Thoughtfully evaluates whether the methodology is appropriate for the research questions and discusses plausible alternative methods.
Findings & Policy Implications	2	Accurately summarizes the main findings and clearly explains their implications for policy and/or practice.
Organization & Delivery	1	Presentation is well-organized, stays within the time limit, and all group members contribute to the presentation.

Questions?

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15-minute quiz
10-minute break